# The Physics Architecture Between the Shielding Calculators

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## Contents

# 1 The Object Being Shadowed

All twelve calculators point at the same physical object, even when they do not compute it. That object is the nuclear magnetic shielding tensor. Place a nucleus in an externally applied magnetic field $\boldsymbol{B}_0$. The electrons move and polarize in response. Their induced motion creates a secondary magnetic field at the nucleus, so the nucleus sees

$$\boldsymbol{B}_{\mathrm{nuc}} = (\boldsymbol{I} - \boldsymbol{\sigma})\boldsymbol{B}_0.$$

$\boldsymbol{B}_{\mathrm{nuc}}$ is the magnetic field at the nucleus. $\boldsymbol{I}$ is the $3 \times 3$ identity matrix. $\boldsymbol{\sigma}$ is the shielding tensor: a dimensionless $3 \times 3$ linear-response coefficient. Its entry $\sigma_{ab}$ says how an applied-field component along direction $b$ reduces or redirects the magnetic field component along direction $a$ at the nucleus. The minus sign is the shielding convention: positive shielding means the induced electronic field screens the applied field.

This definition is already a map. A molecule is not a sphere. If the applied field is turned relative to a peptide plane, an aromatic ring, a hydrogen bond, or a water dipole, the induced electronic current need not point in the same direction as before. A scalar chemical shift is the shadow cast by this tensor after orientational averaging, reference subtraction, and a sign convention. The calculators keep the tensorial shadow before that collapse whenever the code has a rank-2 geometric object to emit.

Ramsey's quantum theory gives the source of the tensor: shielding is a second-order response of the electronic state to a nuclear magnetic moment and an external magnetic field [1, 2]. In the current-density picture, the same response can be written as a Biot–Savart field generated by the current induced by $\boldsymbol{B}_0$:

$$\sigma_{ab}(I) = -\frac{\mu_0}{4\pi} \int \frac{\left[(\partial \boldsymbol{j}(\boldsymbol{x})/\partial B_{0,b}) \times (\boldsymbol{R}_I - \boldsymbol{x})\right]_a}{|\boldsymbol{R}_I - \boldsymbol{x}|^3} \, d^3x.$$

$\boldsymbol{R}_I$ is the position of nucleus $I$. $\boldsymbol{x}$ is an electronic position in space. $\boldsymbol{j}(\boldsymbol{x})$ is the electronic current density. The derivative $\partial \boldsymbol{j}/\partial B_{0,b}$ is the part of that current which changes linearly when the applied field component $B_{0,b}$ is changed. The cross product and inverse-cube denominator are the Biot–Savart kernel: they turn the field-induced part of the current at $\boldsymbol{x}$ into magnetic field at the nucleus. The leading minus sign converts induced field into shielding by the convention above.

None of the geometric calculators evaluates this integral over the true quantum current density. They replace pieces of the electronic response by classical sources: a current loop, a susceptibility axis, a point charge, a multipole, a hydrogen-bond geometry, a dispersion contact, or a solvent shell. The output is therefore a geometric kernel until a calibration step attaches the missing response coefficient. LarsenHBond is the partial exception: its tables already contain DFT shielding-difference tensors in $\mathrm{ppm}$, but those tensors are still local empirical terms, not a complete chemical shift.

# 2 The Shared Tensor Language

The code stores a $3 \times 3$ tensor $\boldsymbol{X}$ by splitting its nine Cartesian entries into three pieces. This is not a diagonalization and not a choice of principal axes. It is a closed-form change of coordinates on the tensor entries.

The first piece is the scalar trace average,

$$T_0 = \frac{1}{3}\,\mathrm{tr}\,\boldsymbol{X} = \frac{X_{xx} + X_{yy} + X_{zz}}{3}.$$

It is the part proportional to the identity matrix. For a true shielding tensor, this is the isotropic shielding.

The second piece is the antisymmetric part, stored as a three-component vector,

$$\boldsymbol{T}_1 = \frac{1}{2}\begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}.$$

It records the transpose-odd part of the tensor. Standard isotropic solution NMR usually does not observe this part directly, but several code kernels are asymmetric, so the representation keeps it.

The third piece is the symmetric traceless matrix

$$\boldsymbol{S} = \frac{\boldsymbol{X} + \boldsymbol{X}^{\top}}{2} - T_0 \boldsymbol{I}, \qquad \operatorname{tr} \boldsymbol{S} = 0.$$

Symmetric means $S_{ab} = S_{ba}$. Traceless means the three diagonal entries sum to zero. Those two conditions leave five independent numbers. The code packs them as

$$\boldsymbol{T}_2 = \begin{pmatrix} \sqrt{2}\, S_{xy} \\ \sqrt{2}\, S_{yz} \\ \sqrt{3/2}\, S_{zz} \\ \sqrt{2}\, S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}.$$

The $\sqrt{2}$ factors on the off-diagonal entries pay for double counting: $S_{xy}$ appears twice in the full symmetric matrix, as $S_{xy}$ and $S_{yx}$. The diagonal factors do the same accounting for the two remaining diagonal degrees of freedom after the trace is removed. With these constants, $\sum_m T_{2m}^2 = \sum_{ab} S_{ab}^2$. The five stored values have the same length as the underlying symmetric traceless matrix measured by the Frobenius norm.

In plain representation terms: the nine components of a $3 \times 3$ tensor split as $1 + 3 + 5$. One number is the trace average. Three numbers are the antisymmetric twist. Five numbers are the pure orientation-dependent stretch and squeeze. The file order used throughout the calculators is

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

## 3 The Kernel That Keeps Returning

The classical kernel at the centre of the architecture is the second derivative of $1/\rho$. Let $\boldsymbol{\rho} = \boldsymbol{r} - \boldsymbol{s}$ be the vector from a source point $\boldsymbol{s}$ to an observation point $\boldsymbol{r}$, let $\rho = |\boldsymbol{\rho}|$, and let $\delta_{ab}$ be one when Cartesian indices $a$ and $b$ are equal and zero otherwise. Then

$$D_{ab}(\boldsymbol{\rho}) = \frac{\partial^2}{\partial \rho_a \partial \rho_b} \frac{1}{\rho} = \frac{3\rho_a \rho_b - \delta_{ab}\rho^2}{\rho^5} = \frac{3\hat{\rho}_a \hat{\rho}_b - \delta_{ab}}{\rho^3},$$

where $\hat{\boldsymbol{\rho}} = \boldsymbol{\rho}/\rho$. This matrix is symmetric and traceless. Along the line of sight it has eigenvalue $+2/\rho^3$. In the two transverse directions it has eigenvalue $-1/\rho^3$. The three eigenvalues sum to zero.

This same object appears with different physical names. In magnetostatics, it is the field shape of a point magnetic dipole:

$$B_a(\boldsymbol{r}) \propto D_{ab}(\boldsymbol{\rho}) m_b,$$

where $m_b$ is component $b$ of the dipole moment. In electrostatics, it is the Hessian of a point-charge potential, and therefore the electric-field gradient away from the charge. In both cases, the shape is the same because both fields descend from $1/\rho$.

If a source has a unit axis $\hat{\boldsymbol{u}}$, the scalar made by looking along that axis is the double contraction

$$f(\boldsymbol{\rho}, \hat{\boldsymbol{u}}) = \hat{u}_a D_{ab}(\boldsymbol{\rho}) \hat{u}_b = \frac{3\cos^2\theta - 1}{\rho^3}, \qquad \cos\theta = \hat{\boldsymbol{\rho}} \cdot \hat{\boldsymbol{u}}.$$

This is the two-lobed dipolar angular law. It is positive near the source axis, negative broadside, and zero when $\cos^2\theta = 1/3$, the magic angle near $54.7°$. McConnell, RingSusceptibility, and HBond all write this scalar as the trace average $T_0$ of their full asymmetric kernel. Dispersion uses the same angular matrix but replaces the inverse-cube distance law with an inverse-sixth-power contact law. WaterField and APBSField use the same second-derivative structure as an electric-field-gradient tensor. Haigh–Mallion integrates it over a ring surface. PiQuadrupole differentiates it two more times.

## 4 One Network

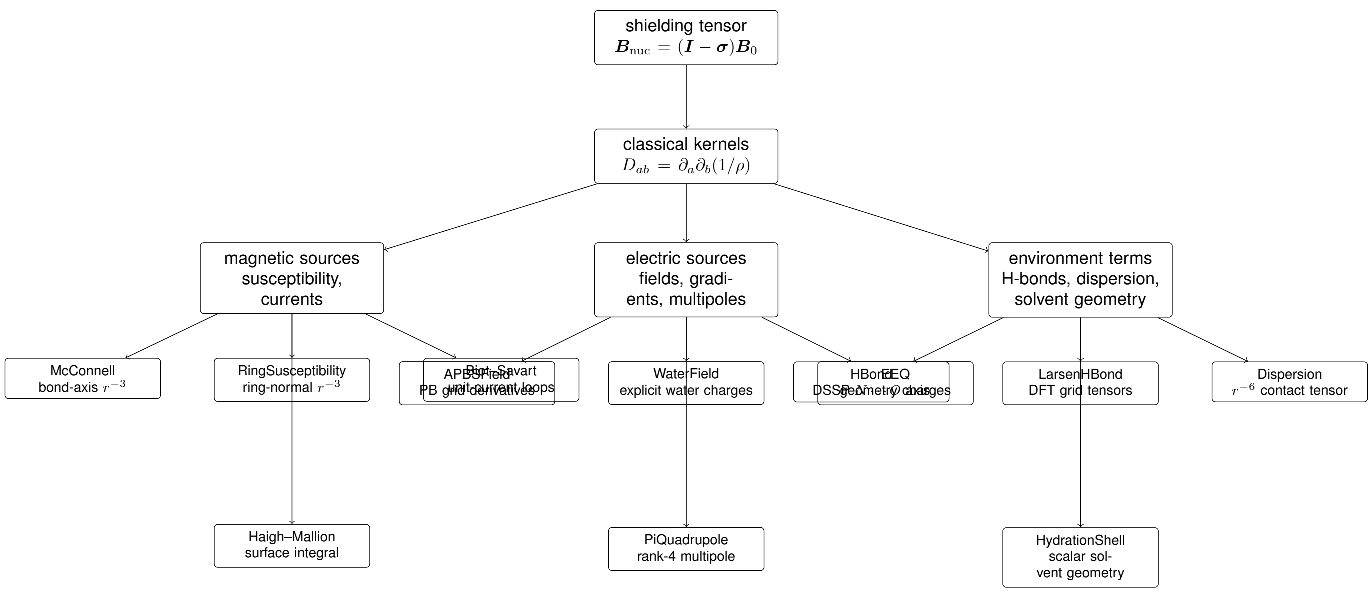


Figure 1: The calculators are not isolated formulas. Most are different classical reductions of the same response problem: electronic currents create shielding, and the code measures geometric kernels that could organize those currents after calibration. The charge model and the solvent-geometry calculator are the exceptions: they supply inputs and context, not a reduction of the response.

The diagram separates source mechanisms, not code ownership. The ring-current calculators live together because they all ask how an aromatic ring responds to $\boldsymbol{B}_0$, but they make different classical replacements for the unknown quantum current. The electrostatic calculators live together because they all read the local potential landscape: first derivatives as fields, second derivatives as field gradients, and higher derivatives as multipoles. The environment calculators sit on the boundary: they do not all emit tensors, but they report the solvent, contact, and hydrogen-bond geometry that can change the electronic response behind $\boldsymbol{\sigma}$.

## 5 Magnetic Susceptibility: Bonds, Rings, and H-Bonds

A small anisotropic magnetic object has a preferred axis. Call that unit axis $\hat{\boldsymbol{u}}$. For a covalent bond, $\hat{\boldsymbol{u}}$ is the bond direction. For RingSusceptibility, it is the aromatic ring normal. For HBond, it is the donor-$N$ to acceptor-$O$ direction supplied after DSSP has identified the backbone hydrogen bond [18]. Let $\boldsymbol{m}$ be the source midpoint or ring centre, let $\boldsymbol{r}_i$ be the target atom position, and define

$$\boldsymbol{d} = \boldsymbol{r}_i - \boldsymbol{m}, \qquad r = |\boldsymbol{d}|, \qquad \hat{\boldsymbol{d}} = \boldsymbol{d}/r, \qquad c = \hat{\boldsymbol{d}} \cdot \hat{\boldsymbol{u}}.$$

$\boldsymbol{d}$ points from the source centre to the target atom. $r$ is the centre-to-atom distance. $c$ is the cosine of the angle between the observation direction and the source axis.

The shared full tensor used by this family is

$$M_{ab} = \frac{9c\,\hat{d}_a\hat{u}_b - 3\hat{u}_a\hat{u}_b - \left(3\hat{d}_a\hat{d}_b - \delta_{ab}\right)}{r^3}.$$

The first term is the dyadic product of the observation direction and the source axis, weighted by the angle $c$. It is generally asymmetric, so it can create $T_1$. The second term projects onto the source axis. The third term is the negative of the dipolar kernel $D_{ab}$. The trace identity is

$$\frac{1}{3}\,\mathrm{tr}\,\boldsymbol{M} = \frac{3c^2 - 1}{r^3}.$$

Thus the familiar McConnell scalar is not a separate law replacing the tensor; in this code it is the $T_0$ component of the full geometric tensor.

McConnell applies this expression to accepted covalent bonds. It collapses each bond to its midpoint and axis, applies distance and near-field exclusions, and sums unit-weight tensors in Å$^{-3}$. It does not multiply by a bond susceptibility anisotropy. It therefore preserves the angular source geometry from McConnell's long-range shielding theory [5], but not the material coefficient that would turn the geometry into physical shielding.

RingSusceptibility makes the substitution $\hat{\boldsymbol{u}} \mapsto \hat{\boldsymbol{n}}$, where $\hat{\boldsymbol{n}}$ is the fitted aromatic ring normal. It is the point-susceptibility version of an aromatic ring: the whole ring becomes a small anisotropic magnetic object at its centre. It shares the same $r^{-3}$ kernel as McConnell, but the axis is a plane normal rather than a bond direction. The code does not use ring-current intensities or Johnson–Bovey lobe offsets in this calculator; every accepted aromatic ring has unit geometric weight. Case's DFT calibration of aromatic ring-current effects is the natural ancestor for the later calibration step, not a coefficient already applied here [8].

HBond makes a different substitution: $\hat{\boldsymbol{u}} \mapsto \hat{\boldsymbol{h}}$, the donor-$N$ to acceptor-$O$ unit direction, with the source point at the $N \cdots O$ midpoint. Its physical premise is not that the hydrogen bond is a literal magnetic susceptibility rod. It is that backbone hydrogen bonding reorganizes electronic structure, and that the $N \cdots O$ line supplies a strong local axis for the rank-2 shielding response. The code uses the same unit-weight McConnell-family tensor as a geometric proxy. It does not multiply by DSSP energy, donor–hydrogen angle, charge, or an empirical shielding coefficient. The broader physical claim that hydrogen-bond networks strongly affect amide-proton shifts is supported by quantum and empirical studies of backbone hydrogen bonding [19, 20].

The shared origin is the dipolar kernel. The divergence is the identity of the axis and the source list. McConnell gets axes from covalent bonds. RingSusceptibility gets axes from aromatic planes. HBond gets axes from a hydrogen-bond assignment. They are the same symbolic machine with different source semantics.

# 6 Aromatic Ring Currents

Aromatic rings are special because they offer a more literal current picture. The applied magnetic field threads the $\pi$-electron system. In a classical model that response is a circulating current, and a current produces magnetic field by the Biot–Savart law. The canonical ring-current literature begins in the pre-digital era and is carried through later reviews and calibrations [4, 6, 7, 8, 9]. The code implements three different reductions of that idea.

BiotSavart keeps the current-loop picture. For a current $I$ flowing along a curve $C$, the magnetic field at a point $\boldsymbol{r}$ is

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\boldsymbol{\ell} \times (\boldsymbol{r} - \boldsymbol{s})}{|\boldsymbol{r} - \boldsymbol{s}|^3}.$$

$d\boldsymbol{\ell}$ is a small directed segment of the current path at source point $\boldsymbol{s}$. The Johnson–Bovey model replaces an aromatic ring by two displaced polygonal loops, one on each side of the fitted ring plane. The code evaluates each straight segment by a closed-form finite-wire expression, sums the two offset polygons with a unit $1\,\mathrm{nA}$ current, and then forms the shielding-like kernel

$$G_{ab} = -10^6\, B_a^{(1)} n_b.$$

$B_a^{(1)}$ is component $a$ of the unit-current magnetic field at the target atom. $n_b$ is component $b$ of the ring normal. The outer product says that only the applied-field component normal to the ring drives this current in the model, and the resulting secondary field can point in any Cartesian direction. The matrix has rank at most one and can be asymmetric. The code records the ring-current intensity for provenance but does not multiply it into the tensor.

Haigh–Mallion keeps the dipole-field kernel and changes the source from a wire to a surface. For a target atom at $\boldsymbol{r}$, a point $s$ on the ring surface, and $\boldsymbol{\rho} = \boldsymbol{r} - \boldsymbol{s}$, the raw surface tensor is

$$H_{ab}(\boldsymbol{r}) = \int_S \left( \frac{3\rho_a \rho_b}{\rho^5} - \frac{\delta_{ab}}{\rho^3} \right) dS.$$

$S$ is the fan-triangulated aromatic ring surface and $dS$ is a surface area element. The integrand is the same $D_{ab}$ kernel. The code then contracts this surface response with the ring normal and forms

$$V_a = H_{ab} n_b, \qquad G_{ab} = -V_a n_b.$$

The raw $\boldsymbol{H}$ is symmetric and traceless in exact arithmetic because it is an integral of symmetric traceless kernels. The reported $\boldsymbol{G}$ is the rank-at-most-one outer product after the normal contraction. It can have $T_0$, $T_1$, and $T_2$. Like BiotSavart, the result is an unscaled geometric kernel.

RingSusceptibility is the point-source limit of the same family. It does not integrate wires or surfaces. It uses the ring normal in the McConnell-family closed form from the previous section. That makes it cheaper and more local, but it carries less information about ring extent and current path.

These three ring calculators therefore disagree in a useful way. BiotSavart asks what field a pair of current loops would produce. Haigh–Mallion asks what field a sheet of dipole kernels over the ring interior would produce. RingSusceptibility asks what an anisotropic point source at the centre would produce. They share an applied-field projection onto the ring normal and a secondary magnetic field at the target atom. They diverge in what they believe the ring current is between those two events.

# 7 Electric Fields, Gradients, and Charges

Electric fields enter shielding because the electron cloud that responds to $\boldsymbol{B}_0$ is also shaped by electrostatic environment. Buckingham's classic expansion writes shielding changes as functions of local electric fields and electric-field gradients [10]. In the code, those response coefficients are outside the calculators. The calculators report the field or gradient that such a coefficient could multiply.

For a point charge $q$ at source point $\boldsymbol{s}$, a target point $\boldsymbol{r}$, and $\boldsymbol{\rho} = \boldsymbol{r} - \boldsymbol{s}$, the potential is proportional to $q/\rho$. Its first derivative gives an electric field up to a sign convention, and its second derivative gives a rank-2 field-gradient tensor:

$$V_{ab} = k_e q \left( \frac{3\rho_a \rho_b}{\rho^5} - \frac{\delta_{ab}}{\rho^3} \right).$$

$k_e$ is the Coulomb conversion constant in the code's $\mathrm{V}$ Å units when charges are in elementary-charge units and distances are in Å. Again the bracket is the dipolar kernel. The tensor is symmetric and traceless away from the charge.

WaterField evaluates this expression directly for explicit solvent waters. A water molecule enters the sum when its oxygen lies inside the configured oxygen cutoff; then all three point-charge sites contribute. The vector channel uses the code's target-to-charge displacement convention, so it is the gradient of the potential with the sign used by WaterField, not necessarily the textbook $-\nabla\phi$ direction. The tensor channel is unaffected by reversing $\boldsymbol{\rho}$ because it contains $\rho_a \rho_b$. After the point-charge sum, the code removes trace residue and writes the five $T_2$ components. This is explicit solvent electrostatics with no continuum reaction field, no Ewald tail, and no induced polarization inside the calculator.

APBSField obtains the same local objects by a different route. APBS solves a linearized Poisson–Boltzmann problem for a solvated protein potential on a grid [11]. The C++ calculator receives only the scalar potential $\phi$, not APBS-internal fields. At atom position $\boldsymbol{r}_i$, with grid spacing $h_a$ along Cartesian axis $a$, it computes

$$E_a(\boldsymbol{r}_i) = -\frac{\phi(\boldsymbol{r}_i + h_a \hat{\boldsymbol{e}}_a) - \phi(\boldsymbol{r}_i - h_a \hat{\boldsymbol{e}}_a)}{2h_a},$$

where $\hat{\boldsymbol{e}}_a$ is the unit vector along axis $a$. The minus sign is the electrostatic convention $\boldsymbol{E} = -\nabla\phi$. It then computes

$$G_{ab}(\boldsymbol{r}_i) = \frac{E_a(\boldsymbol{r}_i + h_b \hat{\boldsymbol{e}}_b) - E_a(\boldsymbol{r}_i - h_b \hat{\boldsymbol{e}}_b)}{2h_b}.$$

$G_{ab}$ is the derivative of field component $a$ with respect to coordinate $b$. The code symmetrizes this finite-difference matrix and subtracts its trace before emitting the $T_2$ field-gradient feature. Compared with WaterField, APBSField trades explicit water molecules for dielectric boundary and ionic screening. Both calculators end at the same shape of object — a local field vector and a symmetric traceless field-gradient tensor — but not at the same sign. WaterField stores the Hessian of the potential, while APBSField builds its gradient from $\boldsymbol{E} = -\nabla\phi$ and so stores the negative Hessian. The tensor structure matches; the field-gradient sign convention does not.

EEQ sits one step upstream. It assigns geometry-dependent partial charges by a constrained quadratic charge-equilibration model. With effective electronegativities $\chi_i^{\mathrm{eff}}$, a symmetric Coulomb/hardness matrix $\boldsymbol{A}$, charges $q_i$, and total charge $Q$, the model minimizes

$$E(\boldsymbol{q}) = \sum_i \chi_i^{\mathrm{eff}} q_i + \frac{1}{2} \sum_{ij} A_{ij} q_i q_j, \qquad \sum_i q_i = Q.$$

The equalization condition is that the marginal charge energy is the same on all atoms at the constrained minimum. The code solves this system by factoring $\boldsymbol{A}$, enforcing the charge sum (before an output clamp that can move the stored sum slightly off $Q$), and writing the scalar charges and smooth coordination numbers. It does not compute an electric field or shielding tensor. The connection is architectural: a charge model is a possible source for the field and gradient kernels used elsewhere. The D4/EEQ lineage supports this kind of geometry-dependent atomic charge model [15, 14].

Protein electric fields are not a decorative addition to shielding. NMR chemical-shift perturbations have been used to probe protein electric fields, and Poisson–Boltzmann or Coulomb fields are directly compared with those perturbations in the literature [12, 13]. The calculators reflect the same route at kernel level: charge distribution to field, field to field-gradient, field-gradient to a possible calibrated shielding response.

## 8 Multipoles

A multipole expansion is a controlled way to say what a localized charge distribution looks like from far away. Around a source centre, the potential can be written schematically as

$$\phi(\boldsymbol{r}) = q\,\frac{1}{r} - p_a\,\partial_a\frac{1}{r} + \frac{1}{2}Q_{ab}\,\partial_a\partial_b\frac{1}{r} + \cdots .$$

$q$ is the monopole charge. $p_a$ is component $a$ of the dipole moment. $Q_{ab}$ is a quadrupole tensor. Repeated Cartesian indices are summed. Each step in the expansion takes another derivative of $1/r$, so each step carries a sharper angular pattern and a shorter distance law. Distributed multipole analysis and multipole shielding-polarizability work provide the formal background for this source language [16, 17].

PiQuadrupole uses the aromatic ring as an axial quadrupole source. The source is placed at the ring centre and aligned with the fitted ring normal $\hat{\boldsymbol{n}}$. For a target displacement $\boldsymbol{d} = \boldsymbol{r}_i - \boldsymbol{c}$, distance $r = |\boldsymbol{d}|$, and height $h = \boldsymbol{d}\cdot\hat{\boldsymbol{n}}$, the code evaluates the pre-contracted rank-4 derivative

$$G_{ab} = T_{abcd}\hat{n}_c\hat{n}_d, \qquad T_{abcd} = \partial_a\partial_b\partial_c\partial_d\frac{1}{r}.$$

The contraction with $\hat{n}_c\hat{n}_d$ is done algebraically before runtime, leaving the $3\times 3$ closed form

$$\begin{aligned} G_{ab} = {} & \frac{105h^2 d_a d_b}{r^9} - \frac{30h(\hat{n}_a d_b + \hat{n}_b d_a)}{r^7} - \frac{15 d_a d_b}{r^7} + \frac{6\hat{n}_a\hat{n}_b}{r^5} \\ & + \delta_{ab}\left(\frac{3}{r^5} - \frac{15h^2}{r^7}\right). \end{aligned}$$

Every completed term has units Å$^{-5}$. The kernel is symmetric and traceless analytically away from the source, so its signal is in $T_2$, apart from floating-point residue kept by the code. The calculator also writes the scalar

$$q_{\text{quad}} = \frac{3(h/r)^2 - 1}{r^4},$$

which has units Å$^{-4}$. That scalar is not the trace of $\boldsymbol{G}$; it is a separate Buckingham-style angular field kernel.

The quadrupole calculator is therefore not a new physical universe. It is the same $1/r$ source law differentiated farther. WaterField and APBSField stop at second derivatives because they want field gradients from charges and potentials. PiQuadrupole uses fourth derivatives because a quadrupole source already contains two source-side derivatives, and an electric-field gradient at the target adds two target-side derivatives. The distance law changes from $r^{-3}$ for dipolar/EFG kernels to $r^{-5}$ for a quadrupole EFG.

## 9 Hydrogen-Bond Tables

LarsenHBond belongs in the same physical architecture as HBond, but not in the same mathematical family. HBond emits a geometric McConnell-family kernel on DSSP-resolved $N\cdots O$ segments. LarsenHBond looks up a DFT-derived shielding-difference tensor from the ProCS15 parameterization [20].

For each candidate donor hydrogen and acceptor oxygen, the lookup geometry is

$$r = \|\boldsymbol{o} - \boldsymbol{h}\|, \qquad \theta = \angle(\mathrm{H-O-C}), \qquad \rho = \mathrm{dihedral}(\mathrm{H-O-C-third}).$$

$\boldsymbol{h}$ is the donor hydrogen position. $\boldsymbol{o}$ is the acceptor oxygen position. $\boldsymbol{c}$ is the heavy atom bonded to the acceptor oxygen, and $\mathrm{third}$ is the next atom that defines the acceptor plane. The runtime uses $r, \theta, \rho$, a donor class, and an acceptor class to interpolate a tensor-valued grid. The grid tensor is stored in a donor-centred canonical frame, so the code builds an orthonormal donor frame and rotates the returned matrix into the protein frame:

$$\boldsymbol{\sigma}_{\mathrm{lab}} = R^{\mathsf{T}} \boldsymbol{\sigma}_{\mathrm{canon}} R.$$

$R$ is the rotation matrix whose rows are the donor-frame axes in protein coordinates. Both tensor indices rotate because shielding is a rank-2 object.

The physical bridge is the same as the HBond bridge: hydrogen bonding polarizes the donor and acceptor electronic structure, changing the shielding response. The mathematical bridge is different. HBond says, "keep the axis and the inverse-cube angular kernel, then calibrate later." LarsenHBond says, "use a precomputed electronic-structure table indexed by local hydrogen-bond geometry, then rotate the tensor into the molecular frame." The first is a geometric kernel. The second is a local empirical shielding term in $\mathrm{ppm}$. Neither is a complete chemical shift by itself.

## 10 Dispersion and Contact Geometry

London dispersion enters from correlated electronic fluctuations, not from a static charge or a permanent current. Its familiar pair-energy distance law is $-C_6/r^6$, with $C_6$ carrying polarizability information [21]. Modern dispersion corrections make the same point in atom-pair form: the coefficient is chemically meaningful and environment dependent [22, 15]. The Dispersion calculator deliberately stops before that coefficient. It sets $C_6 = 1$, restricts sources to aromatic ring vertices, and records an angular contact descriptor.

For a target atom at $\boldsymbol{r}_i$, an aromatic vertex at $\boldsymbol{v}_j$, and $\boldsymbol{d} = \boldsymbol{r}_i - \boldsymbol{v}_j$, the per-vertex tensor is

$$K_{ab} = \frac{S(r)}{r^6}\left(3\hat{d}_a\hat{d}_b - \delta_{ab}\right), \qquad r = |\boldsymbol{d}|.$$

$S(r)$ is the smooth switching function that is one before the switching onset, falls to zero before the cutoff, and is zero at and beyond the cutoff. The tensor is symmetric and traceless, so its exact signal is in $T_2$. Along the vertex-to-atom direction it has eigenvalue $2S/r^6$; in each transverse direction it has eigenvalue $-S/r^6$. The scalar stored beside it is

$$q_{\mathrm{disp}} = \frac{S(r)}{r^6},$$

which is a positive contact magnitude and not the trace of $\boldsymbol{K}$. The emitted channel sums this magnitude over the accepted aromatic vertices; it is not a single per-vertex value.

The angular shape is the familiar dipolar matrix. The distance law is not. A magnetic dipole or electric field gradient falls as $r^{-3}$. This contact falls as $r^{-6}$, because it is representing short-range polarizable contact geometry, not a Maxwell field from a permanent source. Direct links between London–van der Waals interactions and nuclear magnetic shielding have been worked out for simpler systems [24], and modern chemistry reviews emphasize that dispersion contacts are real electronic interactions [23]. The code's claim is narrower: nearby aromatic vertices produce an orientation-resolved $r^{-6}$ descriptor that can be tested against DFT shielding differences after calibration.

## 11 Solvent Geometry Without a Tensor

HydrationShell does not emit a tensor. It belongs in the architecture because the shielding response of a protein atom is not set by the protein coordinates alone. The local electron cloud sees water, ions, and the dielectric boundary. Explicit first-shell waters can affect computed shielding, and protein hydration layers have structured but dynamic water organization [25, 26, 27].

For atom $i$, HydrationShell forms the coordinate centroid of the protein,

$$\boldsymbol{c} = \frac{1}{N}\sum_{j=1}^{N}\boldsymbol{r}_j,$$

then points from the atom toward that centroid,

$$\hat{\boldsymbol{u}}_i = \frac{\boldsymbol{c}-\boldsymbol{r}_i}{|\boldsymbol{c}-\boldsymbol{r}_i|}.$$

Water oxygen positions within the first-shell cutoff enter the shell. A water on the non-centroid side of the plane through the atom perpendicular to $\hat{\boldsymbol{u}}_i$ counts as exposed; a water on the centroid side counts as buried. The written half-shell fraction is

$$h_i = \frac{n_{\text{exposed},i}}{n_{\text{exposed},i} + n_{\text{buried},i}}$$

when the denominator is nonzero, and zero otherwise.

The water orientation channel averages a cosine. For water $w$, the code's water dipole vector is

$$\boldsymbol{\mu}_w = q_{\mathrm{H}}(\boldsymbol{h}_{1w} - \boldsymbol{o}_w) + q_{\mathrm{H}}(\boldsymbol{h}_{2w} - \boldsymbol{o}_w),$$

where $\boldsymbol{o}_w$ is the oxygen position, $\boldsymbol{h}_{1w}$ and $\boldsymbol{h}_{2w}$ are hydrogen positions, and $q_{\mathrm{H}}$ is the topology hydrogen charge. The averaged quantity is

$$d_i = \frac{1}{m_i}\sum_{w}\frac{\boldsymbol{\mu}_w\cdot\hat{\boldsymbol{r}}_{iw}}{|\boldsymbol{\mu}_w|}, \qquad \hat{\boldsymbol{r}}_{iw} = \frac{\boldsymbol{o}_w-\boldsymbol{r}_i}{|\boldsymbol{o}_w-\boldsymbol{r}_i|},$$

over first-shell waters with nonzero dipole magnitude. The calculator also writes the nearest-ion distance inside a cutoff and the nearest-ion charge.

These four numbers are not field derivatives and not shielding components. They are solvent-structure descriptors. Their place in the symbolic architecture is contextual: they report which side of the atom has water, how those waters are oriented, and whether an ion is nearby. They can help explain shielding variation without pretending to be a tensor kernel.

## 12 What the Architecture Proves

The calculators do not prove that any raw feature is a shielding value. The code proves a more limited and cleaner statement: for each conformation, it can emit geometry-derived objects with known transformation behaviour, known units, known distance laws, and known omissions. Calibration against DFT shielding tensors is the step that can test whether those objects correlate with shielding contributions and can attach physical scale [3].

The magnetic family is united by the dipolar kernel and by the idea that an applied field induces a source whose axis is known geometrically. McConnell, RingSusceptibility, and HBond use the same closed-form asymmetric tensor with different axes and source lists. BiotSavart and Haigh–Mallion make the aromatic source less pointlike: one integrates current segments, the other integrates a dipolar surface. All of them remain geometric until a ring-current intensity, susceptibility, or fitted coefficient is applied.

The electrostatic family is united by derivatives of a potential. WaterField uses analytic point-charge derivatives from explicit solvent. APBSField finite-differences a Poisson–Boltzmann potential grid. EEQ supplies a possible charge distribution but stops before field construction. PiQuadrupole is the higher-multipole continuation of the same derivative hierarchy.

The environment family supplies what the kernels alone do not: hydrogen-bond geometry that has already been parameterized by DFT in LarsenHBond, short-range polarizable contact geometry in Dispersion, and explicit solvent arrangement in HydrationShell. These are not interchangeable with the field calculators. They add different physical facts about the electronic environment.

The network is therefore a set of shadows cast by one hidden object. The hidden object is the electronic current response in Ramsey shielding. The shadows are dipolar tensors, line integrals, surface integrals, potential derivatives, multipole contractions, table-rotated DFT tensors, contact anisotropies, and solvent geometry. The architecture is useful precisely because the shadows are not all the same. Where they agree, they mark a shared physical origin. Where they disagree, calibration can ask which approximation carries the tensor signal in the DFT reference.

## 13 Reference Status

The cited modern support papers were checked against Crossref DOI metadata. The claims carried by de Dios 1993, Case 1995, Boyd and Skrynnikov 2002, Baker 2001, Hass 2008, Kukic 2013, Caldeweyher 2019, Stone 2005, Kjaer 2011, Parker 2006, Larsen 2015, Grimme 2010, Wagner and Schreiner 2015, Nakasako 2004, Halle 2004, and Martin and Matyushov 2015 were also checked against Europe PMC abstract text. Kromhout and Linder 1969 was checked against the ScienceDirect publisher abstract. Facelli 2011 was checked by DOI metadata and open PMC full text; PubMed/Europe PMC report no abstract.

The pre-digital canon used as named lineage has verified Crossref DOI metadata, but the required abstract source was absent or incomplete during this pass: Ramsey 1950, Pople 1956, McConnell 1957, Johnson and Bovey 1958, Haigh and Mallion 1979, Buckingham 1960, London 1937, and Kabsch and Sander 1983. Some of these records expose Crossref/publisher abstracts, but they are treated here as flagged canon carried through the accessible reviews, restatements, and modern applications cited above.

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